

Chapter 110

Metadata Nodes

This chapter details the Metadata nodes available in Fusion.

The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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Copy Metadata [Meta]



The Copy Metadata node

Copy Metadata Node Introduction

Copy Metadata combines, replaces, or clears the metadata in your image. Metadata can be viewed in a subview of the viewer.

Inputs

The two inputs on the Copy Metadata node are used to connect two 2D images.

Background Input: The orange background input is used for the primary 2D image that is output from the node.

Foreground Input: The green foreground input is used for the secondary 2D image that contains metadata you want merge or overwrite onto the background image.

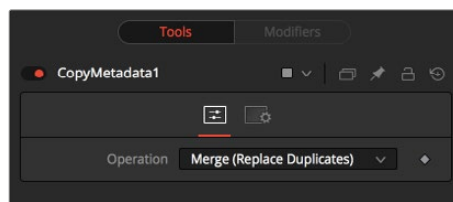
Basic Node Setup

The Copy Metadata node takes metadata from the foreground input (green) and copies it into the background input (orange). The output is the background input with modified metadata.



A Copy Metadata node copies metadata from the foreground and embeds it into the background clip

Inspector



The Copy Metadata Controls tab

Controls Tab

The Controls tab configures how metadata coming from the foreground input image gets added to the background input image.

Operation

The Operation menu determines how the metadata of the foreground and background inputs are treated.

- **Merge (Replace Duplicates):** All values are merged, but values with duplicate names are taken from the foreground input.
- **Merge (Preserve Duplicates):** All values are merged, but values with duplicate names are taken from the background input.
- **Replace:** The metadata in the foreground replaces the entire metadata in the background.
- **Clear:** All metadata is discarded.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Metadata nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Set Metadata [SMeta]



The Set Metadata node

Set Metadata Node Introduction

Set Metadata allows you to create new Name = Value pairs in the metadata. Metadata can be viewed in a subview of the viewer.

Inputs

The single input on the Set Metadata node is used to connect a 2D image that gets metadata added.

Background Input: The orange background input is used for the primary 2D image that is output from the node with the new metadata.

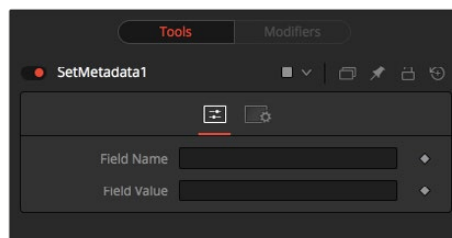
Basic Node Setup

The Set Metadata node embeds new metadata into the background input (orange). The output is the background input with new metadata.



A Set Metadata node creates new metadata and embeds it into the background clip.

Inspector



The Set Metadata Controls tab

Controls Tab

The Controls tab is where you set up the name of the metadata field and the value or information regarding the metadata.

Field Name

The name of the metadata value. Do not use spaces.

Field Value

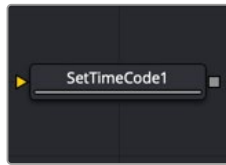
The value assigned to the name above.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Metadata nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Set Timecode [TcMeta]



The Set Timecode node

Set Timecode Node Introduction

Set Timecode inserts dynamic timecode values into the metadata table based on the FPS settings.

Inputs

The single input on the Set Timecode node is used to connect a 2D image that gets timecode added.

Background Input: The orange background input is used for the primary 2D image that is output from the node with the new timecode.

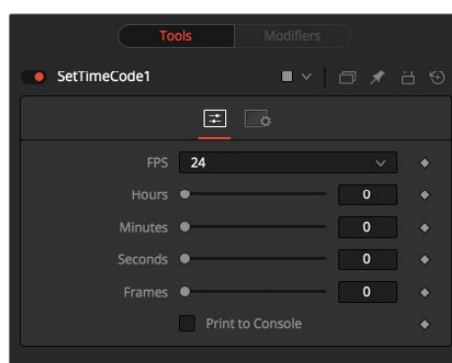
Basic Node Setup

The Set Timecode node embeds new timecode metadata into the background input (orange). The output is the background input with updated timecode.



A Set Timecode node inserts new timecode metadata into the background clip.

Inspector



The Set Timecode Controls tab

Controls Tab

The Controls tab sets the clip's starting timecode metadata based on FPS, hours, minutes, seconds, and frames.

FPS

You can choose from a variety of settings for frames per second.

Since this is a Fuse, you can easily adapt the settings to your needs by editing the appropriate piece of code for the buttons:

```
MBTNC_StretchToFit = true,
{ MBTNC_AddButton = "24" },
{ MBTNC_AddButton = "25" },
{ MBTNC_AddButton = "30" },
{ MBTNC_AddButton = "48" },
{ MBTNC_AddButton = "50" },
{ MBTNC_AddButton = "60" },
})
```

as well as for the actual values:

```
local rates = { 24, 25, 30, 48, 50, 60 }
```

Hours/Minutes/Seconds/Frames Sliders

Define an offset from the starting frame of the current comp.

Print to Console

Verbose output of the Timecode/Frame value in the Console.

The Timecode/Frames conversion is done according to the FPS settings.

The result might look like this:

```
TimeCode: 00:00:08:15
Frames:   207
```

Common Controls

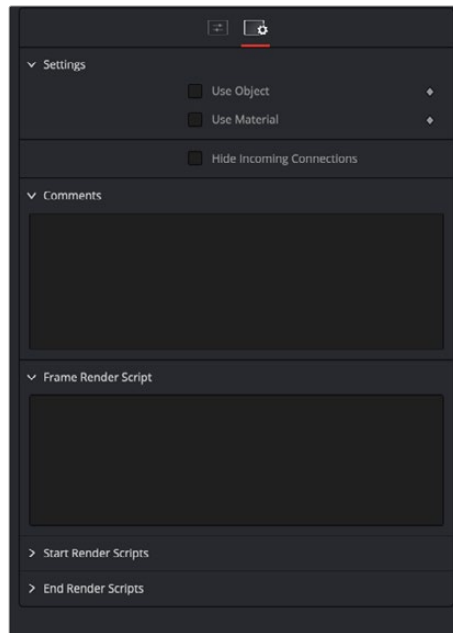
Settings Tab

The Settings tab in the Inspector is also duplicated in other Metadata nodes. These common controls are described in detail in the following “The Common Controls” section.

The Common Controls

Nodes that handle metadata operations share several identical controls in the Inspector. This section describes controls that are common among Metadata nodes.

Inspector



The Common Metadata Settings tab

Settings Tab

The Settings tab in the Inspector can be found on every tool in the Metadata category. The controls are consistent and work the same way for each tool.

Use Object/Use Material (Checkboxes)

Some 3D software can render to file formats that support additional channels. Notably, the EXR file format supports Object ID and material ID channels, which can be used as a mask for the effect. These checkboxes determine whether the channels are used, if present. The specific Material ID or Object ID affected is chosen using the next set of controls.

Correct Edges

This checkbox appears only when the Use Object or Use Material checkboxes are selected. It toggles the method used to deal with overlapping edges of objects in a multi-object image. When enabled, the Coverage and Background Color channels are used to separate and improve the effect around the edge of the object. If this option disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

For more information on the Coverage and Background Color channels, see *Chapter 78, "Understanding Image Channels,"* in the DaVinci Resolve Reference Manual, or Chapter 18 in the Fusion Reference Manual.

Object ID/Material ID (Sliders)

Use these sliders to select which ID is used to create a mask from the object or material channels of an image. Use the Sample button in the same way as the Color Picker: to grab IDs from the image displayed in the view. The image or sequence must have been rendered from a 3D software package with those channels included.

Comments

The Comments field is used to add notes to a tool. Click in the empty field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.